Process and Operations Management
Tutorial 3: Managing Risks

Stefanos Poulidis

Decision Sciences, INSEAD

POM Tutorials

- Session 1: Process Improvement
- Session 2: Managing Variability
- **Session 3: Managing Risks (The Newsvendor Model) I**
- Session 4: Managing Risks (The Newsvendor Model) II

Managing Risks
Step 1: Collect all the inputs c, r, s, μ, σ

Demand parameters

- How is demand distributed?
  - (usually) normal $N(\mu, \sigma^2)$ or discrete
  - $\boldsymbol{\mu}$: mean (average)
  - $\boldsymbol{\sigma}$: standard deviation (or variance, which is σ^2)

Product parameters

- ***c***: unit manufacturing cost

(E.g., the cost of baking a pie is c = $10. This includes your grandma's labor cost, ingredients' cost, etc.)

- ***r***: selling price per unit

(The price at which your grandma sells her pies to her neighbors, e.g. r = $12)

- ***s***: salvage value per unit

(Your grandma can give the pie to you at s = $1 if no one buys it)

Note: usually s < c < r. When the good is perishable usually s = 0.

Managing Risks
Step 2: Calculate underage and overage costs C_u, C_o

- $C_u = r - c$: consequence of producing (stocking) not enough items
  - Profit
  - Opportunity cost

- $C_o = c - s$: consequence of producing (stocking) too many items

Managing Risks
Step 3: Calculate critical ratio

$$\frac{C_u}{C_u + C_o} = \frac{r - c}{r - c + c - s} = \frac{r - c}{r - s}$$

Managing Risks
Step 4: Find order quantity Q^* which maximizes expected profit

Case 1: demand is normal $N(\mu, \sigma^2)$.

– Find z-value from Table 1 (standard normal distribution $\Phi(z)$), i.e. find the z^* such that:

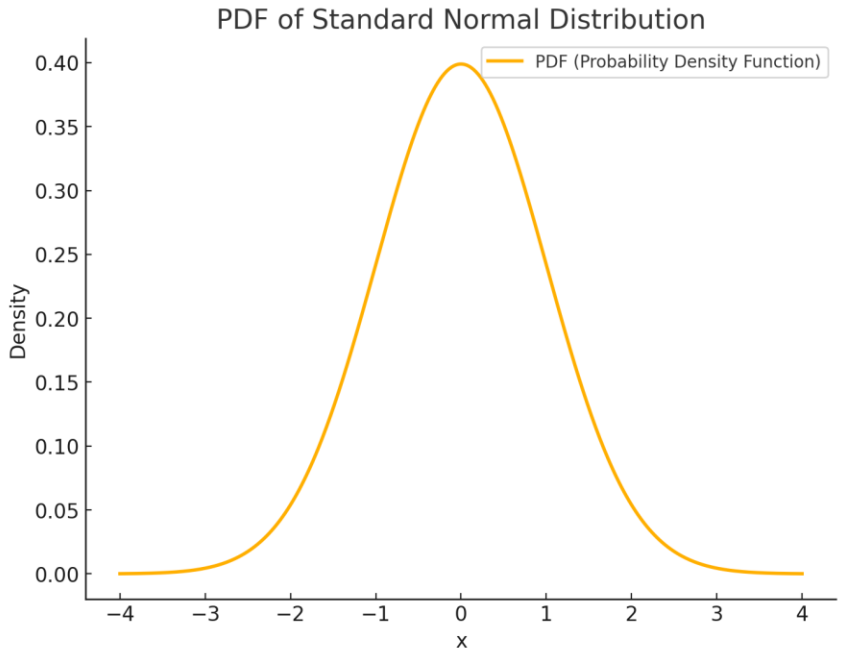
$$\Phi(z^*) = \frac{C_u}{C_u + C_o}$$

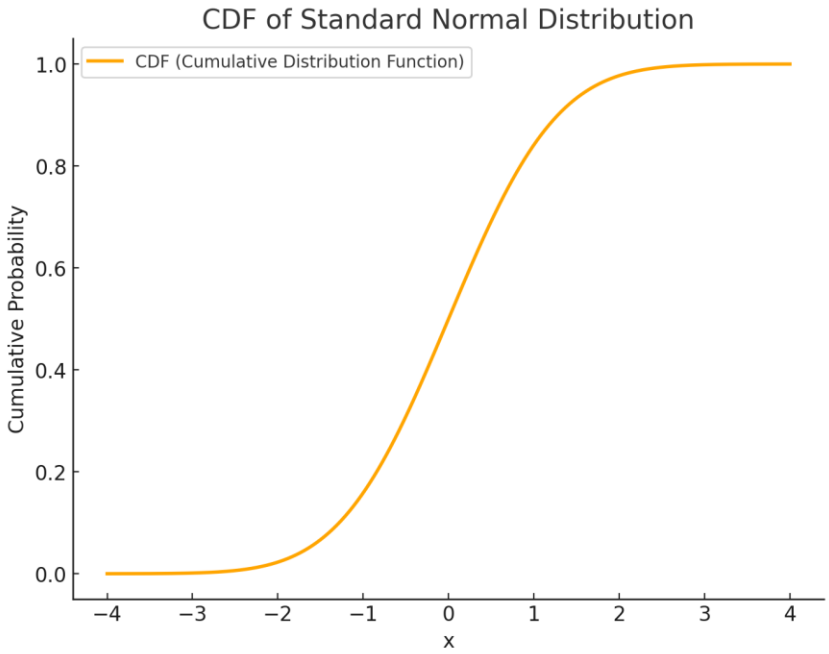
Note: basically you need to find your critical ratio in Table 1 and find the corresponding z-value. If the critical ratio is not there, approximate, taking the closest corresponding z-value in Table 1.

– Calculate Q^* using the formula: $Q^* = \mu + z^* \cdot \sigma$

Note: choose the closest Q^* from above.

Normal distribution





Hundredths decimal point of z

Tenths decimal point of z

z	0.00	-0.01	-0.02	-0.03	-0.04	-0.05	-0.06	-0.07	-0.08	-0.09
-2.9	0.00187	0.00181	0.00175	0.00169	0.00164	0.00159	0.00154	0.00149	0.00144	0.00139
-2.8	0.00256	0.00248	0.00240	0.00233	0.00226	0.00219	0.00212	0.00205	0.00199	0.00193
-2.7	0.00347	0.00336	0.00326	0.00317	0.00307	0.00298	0.00289	0.00280	0.00272	0.00264
-2.6	0.00466	0.00453	0.00440	0.00427	0.00415	0.00402	0.00391	0.00379	0.00368	0.00357
-2.5	0.00621	0.00604	0.00587	0.00570	0.00554	0.00539	0.00523	0.00508	0.00494	0.00480
-2.4	0.00820	0.00798	0.00776	0.00755	0.00734	0.00714	0.00695	0.00676	0.00657	0.00639
-2.3	0.01072	0.01044	0.01017	0.00990	0.00964	0.00939	0.00914	0.00889	0.00866	0.00842
-2.2	0.01390	0.01355	0.01321	0.01287	0.01255	0.01222	0.01191	0.01160	0.01130	0.01101
-2.1	0.01786	0.01743	0.01700	0.01659	0.01618	0.01578	0.01539	0.01500	0.01463	0.01426
-2.0	0.02275	0.02222	0.02169	0.02118	0.02068	0.02018	0.01970	0.01923	0.01876	0.01831
-1.9	0.02872	0.02807	0.02743	0.02680	0.02619	0.02559	0.02500	0.02442	0.02385	0.02330
-1.8	0.03593	0.03515	0.03438	0.03362	0.03288	0.03216	0.03144	0.03074	0.03005	0.02938
-1.7	0.04457	0.04363	0.04272	0.04182	0.04093	0.04006	0.03920	0.03836	0.03754	0.03673
-1.6	0.05480	0.05370	0.05262	0.05155	0.05050	0.04947	0.04846	0.04746	0.04648	0.04551
-1.5	0.06681	0.06552	0.06426	0.06301	0.06178	0.06057	0.05938	0.05821	0.05705	0.05592
-1.4	0.08076	0.07927	0.07780	0.07636	0.07493	0.07353	0.07215	0.07078	0.06944	0.06811
-1.3	0.09680	0.09510	0.09342	0.09176	0.09012	0.08851	0.08691	0.08534	0.08379	0.08226
-1.2	0.11507	0.11314	0.11123	0.10935	0.10749	0.10565	0.10383	0.10204	0.10027	0.09853
-1.1	0.13567	0.13350	0.13136	0.12924	0.12714	0.12507	0.12302	0.12100	0.11900	0.11702
-1.0	0.15866	0.15625	0.15386	0.15151	0.14917	0.14686	0.14457	0.14231	0.14007	0.13786
-0.9	0.18406	0.18141	0.17879	0.17619	0.17361	0.17106	0.16853	0.16602	0.16354	0.16109
-0.8	0.21186	0.20897	0.20611	0.20327	0.20045	0.19766	0.19489	0.19215	0.18943	0.18673
-0.7	0.24196	0.23885	0.23576	0.23270	0.22965	0.22663	0.22363	0.22065	0.21770	0.21476
-0.6	0.27425	0.27093	0.26763	0.26435	0.26109	0.25785	0.25463	0.25143	0.24825	0.24510
-0.5	0.30854	0.30503	0.30153	0.29806	0.29460	0.29116	0.28774	0.28434	0.28096	0.27760
-0.4	0.34458	0.34090	0.33724	0.33360	0.32997	0.32636	0.32276	0.31918	0.31561	0.31207
-0.3	0.38209	0.37828	0.37448	0.37070	0.36693	0.36317	0.35942	0.35569	0.35197	0.34827
-0.2	0.42074	0.41683	0.41294	0.40905	0.40517	0.40129	0.39743	0.39358	0.38974	0.38591
-0.1	0.46017	0.45620	0.45224	0.44828	0.44433	0.44038	0.43644	0.43251	0.42858	0.42465
0.0	0.50000	0.49601	0.49202	0.48803	0.48405	0.48006	0.47608	0.47210	0.46812	0.46414
z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.50000	0.50399	0.50798	0.51197	0.51595	0.51994	0.52392	0.52790	0.53188	0.53586
0.1	0.53983	0.54380	0.54776	0.55172	0.55567	0.55962	0.56356	0.56749	0.57142	0.57535
0.2	0.57926	0.58317	0.58706	0.59095	0.59483	0.59871	0.60257	0.60642	0.61026	0.61409
0.3	0.61791	0.62172	0.62552	0.62930	0.63307	0.63683	0.64058	0.64431	0.64803	0.65173
0.4	0.65542	0.65910	0.66276	0.66640	0.67003	0.67364	0.67724	0.68082	0.68439	0.68793
0.5	0.69146	0.69497	0.69847	0.70194	0.70540	0.70884	0.71226	0.71566	0.71904	0.72240
0.6	0.72575	0.72907	0.73237	0.73565	0.73891	0.74215	0.74537	0.74857	0.75175	0.75490
0.7	0.75804	0.76115	0.76424	0.76730	0.77035	0.77337	0.77637	0.77935	0.78230	0.78524
0.8	0.78814	0.79103	0.79389	0.79673	0.79955	0.80234	0.80511	0.80785	0.81057	0.81327
0.9	0.81594	0.81859	0.82121	0.82381	0.82639	0.82894	0.83147	0.83398	0.83646	0.83891
1.0	0.84134	0.84375	0.84614	0.84849	0.85083	0.85314	0.85543	0.85769	0.85993	0.86214
1.1	0.86433	0.86650	0.86864	0.87076	0.87286	0.87493	0.87698	0.87900	0.88100	0.88298
1.2	0.88493	0.88686	0.88877	0.89065	0.89251	0.89435	0.89617	0.89796	0.89973	0.90147
1.3	0.90320	0.90490	0.90658	0.90824	0.90988	0.91149	0.91309	0.91466	0.91621	0.91774
1.4	0.91924	0.92073	0.92220	0.92364	0.92507	0.92647	0.92785	0.92922	0.93056	0.93189
1.5	0.93319	0.93448	0.93574	0.93699	0.93822	0.93943	0.94062	0.94179	0.94295	0.94408
1.6	0.94520	0.94630	0.94738	0.94845	0.94950	0.95053	0.95154	0.95254	0.95352	0.95449
1.7	0.95543	0.95637	0.95728	0.95818	0.95907	0.95994	0.96080	0.96164	0.96246	0.96327
1.8	0.96407	0.96485	0.96562	0.96638	0.96712	0.96784	0.96856	0.96926	0.96995	0.97062
1.9	0.97128	0.97193	0.97257	0.97320	0.97381	0.97441	0.97500	0.97558	0.97615	0.97670
2.0	0.97725	0.97778	0.97831	0.97882	0.97932	0.97982	0.98030	0.98077	0.98124	0.98169
2.1	0.98214	0.98257	0.98300	0.98341	0.98382	0.98422	0.98461	0.98500	0.98537	0.98574
2.2	0.98610	0.98645	0.98679	0.98713	0.98745	0.98778	0.98809	0.98840	0.98870	0.98899
2.3	0.98928	0.98956	0.98983	0.99010	0.99036	0.99061	0.99086	0.99111	0.99134	0.99158
2.4	0.99180	0.99202	0.99224	0.99245	0.99266	0.99286	0.99305	0.99324	0.99343	0.99361
2.5	0.99379	0.99396	0.99413	0.99430	0.99446	0.99461	0.99477	0.99492	0.99506	0.99520
2.6	0.99534	0.99547	0.99560	0.99573	0.99585	0.99598	0.99609	0.99621	0.99632	0.99643
2.7	0.99653	0.99664	0.99674	0.99683	0.99693	0.99702	0.99711	0.99720	0.99728	0.99736
2.8	0.99744	0.99752	0.99760	0.99767	0.99774	0.99781	0.99788	0.99795	0.99801	0.99807
2.9	0.99813	0.99819	0.99825	0.99831	0.99836	0.99841	0.99846	0.99851	0.99856	0.99861

Values of the critical ratio < 0.5

Values of the critical ratio > 0.5

Managing Risks
Step 4: Find order quantity Q^* which maximizes expected profit

Case 2: demand follows some discrete distribution F(·).

– Find quantity Q^* such that

$$F(Q^*) = \frac{C_u}{C_u + C_o}$$

Note: choose the closest Q^* from above.

Managing Risks
Step 5: Calculate performance measures

- **Expected lost sales** = $\sigma \cdot L(z)$, where $L(z)$ is found from Table 2 of the loss function for standard normal distribution. Note, that z in $L(z)$ is the z^* which you found in Step 3.
- **Expected sales** = μ − Expected lost sales
- **Expected leftover inventory** = Q^* − Expected sales
- **Expected profit** = $C_u \cdot$ Expected sales − $C_o \cdot$ Expected leftover inventory
- **In-stock probability** = $F(Q^*)$: probability that the firm will end up selling season without shortages.
  - In other words, it is the probability that the good will be available (in stock) for every customer.
  - Basically, it is the probability that the realized demand will be less than Q^* that you stock.
- **Stock-out probability** = $1 - F(Q^*)$: probability that there will be shortages during the selling period.

Note: expected leftover inventory and expected lost sales may be positive simultaneously.

Managing Risks

Can expected lost sales and expected leftover inventory both be positive?

Yes.

	Day 1	Day 2	Day 3	Expectation
Demand	6	21	15	
Order	15	15	15	
Lost sales	0	6	0	6/3=2
Leftover inventory	9	0	0	9/3=3

z	0.00	-0.01	-0.02	-0.03	-0.04	-0.05	-0.06	-0.07	-0.08	-0.09
-2.9	2.90054	2.91052	2.92051	2.93049	2.94047	2.95046	2.96044	2.97042	2.98041	2.99040
-2.8	2.80076	2.81074	2.82071	2.83069	2.84066	2.85064	2.86062	2.87060	2.88058	2.89056
-2.7	2.70106	2.71103	2.72099	2.73096	2.74093	2.75090	2.76087	2.77084	2.78081	2.79079
-2.6	2.60146	2.61142	2.62137	2.63133	2.64129	2.65125	2.66121	2.67117	2.68113	2.69110
-2.5	2.50200	2.51194	2.52188	2.53183	2.54177	2.55171	2.56166	2.57161	2.58156	2.59151
-2.4	2.40272	2.41264	2.42256	2.43248	2.44241	2.45234	2.46227	2.47220	2.48213	2.49207
-2.3	2.30366	2.31356	2.32345	2.33335	2.34325	2.35316	2.36307	2.37298	2.38289	2.39280
-2.2	2.20489	2.21475	2.22462	2.23449	2.24436	2.25423	2.26411	2.27400	2.28388	2.29377
-2.1	2.10647	2.11629	2.12612	2.13595	2.14579	2.15563	2.16547	2.17532	2.18517	2.19503
-2.0	2.00849	2.01827	2.02805	2.03783	2.04762	2.05742	2.06722	2.07702	2.08683	2.09665
-1.9	1.91105	1.92077	1.93049	1.94022	1.94996	1.95970	1.96945	1.97920	1.98896	1.99872
-1.8	1.81428	1.82392	1.83357	1.84323	1.85290	1.86257	1.87226	1.88195	1.89164	1.90134
-1.7	1.71829	1.72785	1.73742	1.74699	1.75658	1.76617	1.77578	1.78539	1.79501	1.80464
-1.6	1.62324	1.63270	1.64217	1.65165	1.66114	1.67064	1.68015	1.68967	1.69920	1.70874
-1.5	1.52931	1.53865	1.54800	1.55736	1.56674	1.57612	1.58552	1.59494	1.60436	1.61380
-1.4	1.43667	1.44587	1.45508	1.46431	1.47356	1.48281	1.49208	1.50137	1.51067	1.51998
-1.3	1.34553	1.35457	1.36363	1.37270	1.38179	1.39090	1.40002	1.40916	1.41831	1.42748
-1.2	1.25610	1.26496	1.27384	1.28274	1.29165	1.30059	1.30954	1.31851	1.32750	1.33650
-1.1	1.16862	1.17727	1.18595	1.19465	1.20336	1.21210	1.22086	1.22964	1.23844	1.24726
-1.0	1.08332	1.09174	1.10019	1.10866	1.11716	1.12568	1.13422	1.14279	1.15138	1.15999
-0.9	1.00043	1.00860	1.01680	1.02503	1.03328	1.04156	1.04986	1.05819	1.06654	1.07491
-0.8	0.92021	0.92810	0.93603	0.94398	0.95196	0.95997	0.96801	0.97607	0.98417	0.99229
-0.7	0.84288	0.85048	0.85810	0.86576	0.87345	0.88117	0.88892	0.89669	0.90450	0.91234
-0.6	0.76867	0.77595	0.78325	0.79059	0.79797	0.80537	0.81281	0.82028	0.82778	0.83531
-0.5	0.69780	0.70473	0.71170	0.71870	0.72573	0.73281	0.73991	0.74705	0.75422	0.76143
-0.4	0.63044	0.63701	0.64362	0.65027	0.65695	0.66367	0.67042	0.67721	0.68404	0.69090
-0.3	0.56676	0.57296	0.57920	0.58547	0.59178	0.59813	0.60452	0.61094	0.61740	0.62390
-0.2	0.50689	0.51271	0.51856	0.52445	0.53038	0.53634	0.54235	0.54840	0.55448	0.56060
-0.1	0.45094	0.45635	0.46181	0.46731	0.47285	0.47842	0.48404	0.48969	0.49539	0.50112
0.0	0.39894	0.40396	0.40902	0.41412	0.41926	0.42444	0.42966	0.43492	0.44022	0.44556
z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.39894	0.39396	0.38902	0.38412	0.37926	0.37444	0.36966	0.36492	0.36022	0.35556
0.1	0.35094	0.34635	0.34181	0.33731	0.33285	0.32842	0.32404	0.31969	0.31539	0.31112
0.2	0.30689	0.30271	0.29856	0.29445	0.29038	0.28634	0.28235	0.27840	0.27448	0.27060
0.3	0.26676	0.26296	0.25920	0.25547	0.25178	0.24813	0.24452	0.24094	0.23740	0.23390
0.4	0.23044	0.22701	0.22362	0.22027	0.21695	0.21367	0.21042	0.20721	0.20404	0.20090
0.5	0.19780	0.19473	0.19170	0.18870	0.18573	0.18281	0.17991	0.17705	0.17422	0.17143
0.6	0.16867	0.16595	0.16325	0.16059	0.15797	0.15537	0.15281	0.15028	0.14778	0.14531
0.7	0.14288	0.14048	0.13810	0.13576	0.13345	0.13117	0.12892	0.12669	0.12450	0.12234
0.8	0.12021	0.11810	0.11603	0.11398	0.11196	0.10997	0.10801	0.10607	0.10417	0.10229
0.9	0.10043	0.09860	0.09680	0.09503	0.09328	0.09156	0.08986	0.08819	0.08654	0.08491
1.0	0.08332	0.08174	0.08019	0.07866	0.07716	0.07568	0.07422	0.07279	0.07138	0.06999
1.1	0.06862	0.06727	0.06595	0.06465	0.06336	0.06210	0.06086	0.05964	0.05844	0.05726
1.2	0.05610	0.05496	0.05384	0.05274	0.05165	0.05059	0.04954	0.04851	0.04750	0.04650
1.3	0.04553	0.04457	0.04363	0.04270	0.04179	0.04090	0.04002	0.03916	0.03831	0.03748
1.4	0.03667	0.03587	0.03508	0.03431	0.03356	0.03281	0.03208	0.03137	0.03067	0.02998
1.5	0.02931	0.02865	0.02800	0.02736	0.02674	0.02612	0.02552	0.02494	0.02436	0.02380
1.6	0.02324	0.02270	0.02217	0.02165	0.02114	0.02064	0.02015	0.01967	0.01920	0.01874
1.7	0.01829	0.01785	0.01742	0.01699	0.01658	0.01617	0.01578	0.01539	0.01501	0.01464
1.8	0.01428	0.01392	0.01357	0.01323	0.01290	0.01257	0.01226	0.01195	0.01164	0.01134
1.9	0.01105	0.01077	0.01049	0.01022	0.00996	0.00970	0.00945	0.00920	0.00896	0.00872
2.0	0.00849	0.00827	0.00805	0.00783	0.00762	0.00742	0.00722	0.00702	0.00683	0.00665
2.1	0.00647	0.00629	0.00612	0.00595	0.00579	0.00563	0.00547	0.00532	0.00517	0.00503
2.2	0.00489	0.00475	0.00462	0.00449	0.00436	0.00423	0.00411	0.00400	0.00388	0.00377
2.3	0.00366	0.00356	0.00345	0.00335	0.00325	0.00316	0.00307	0.00298	0.00289	0.00280
2.4	0.00272	0.00264	0.00256	0.00248	0.00241	0.00234	0.00227	0.00220	0.00213	0.00207
2.5	0.00200	0.00194	0.00188	0.00183	0.00177	0.00171	0.00166	0.00161	0.00156	0.00151
2.6	0.00146	0.00142	0.00137	0.00133	0.00129	0.00125	0.00121	0.00117	0.00113	0.00110
2.7	0.00106	0.00103	0.00099	0.00096	0.00093	0.00090	0.00087	0.00084	0.00081	0.00079
2.8	0.00076	0.00074	0.00071	0.00069	0.00066	0.00064	0.00062	0.00060	0.00058	0.00056
2.9	0.00054	0.00052	0.00051	0.00049	0.00047	0.00046	0.00044	0.00042	0.00041	0.00040

Table 2: Standard normal loss function, $L(z)$.

Problem 1: Matt Herron

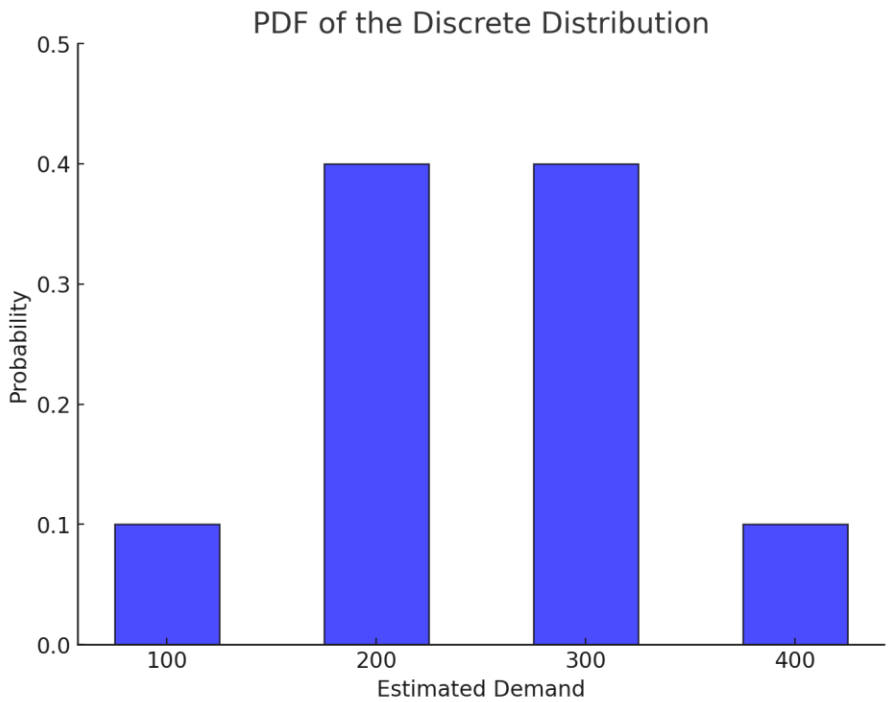
Matt Herron is the chief buyer at Investment Clothiers, a retail store known for excellence in apparel. It is time to order merchandise for the Christmas season. During a recent trip to Hong Kong, Matt spotted a particular men's overcoat that he expects will sell very well. Based on past experience, Matt expects demand to range from 100 to 400 overcoats, with probabilities that are as follows:

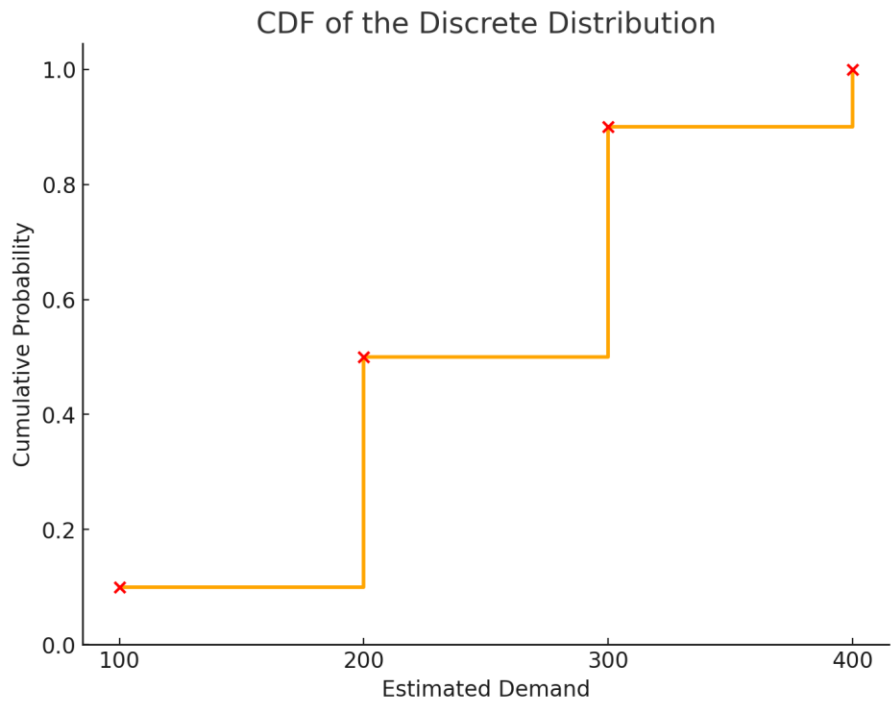
Estimated demand	Probability
100	0.1
200	0.4
300	0.4
400	0.1

The total cost to Investment Clothiers would be $60 per overcoat, and the retail price is estimated at $110 per overcoat. Any overcoat left over after the Christmas season is expected to be sold at $40 each.

Problem 1: Matt Herron

Estimated demand	Probability
100	0.1
200	0.4
300	0.4
400	0.1





Problem 1: Matt Herron

Estimated demand	Probability
100	0.1
200	0.4
300	0.4
400	0.1

(1a) Calculate the expected profits for each of the following four cases: when Matt buys 100, 200, 300 or 400 overcoats. How many overcoats should Matt buy if he wishes to maximize expected profits over the coming Christmas season?

Q = 100: He orders 100, sells them all, as demand is always greater than or equal to his order.

E [profit] = E [revenue] − E [cost] = 100 · 110 − 100 · 60 = $5, 000.

Q = 200: He orders 200, has a 0.10 probability of having an excess stock at the end of the Christmas season of 100 (when D = 100), and sells them at a discount; with probability 0.90 he sells them all.

E [cost] = 200 · 60 = $12, 000.

E [revenue] = 0.10 · (100 · 110 + 100 · 40) + 0.90 · (200 · 110) = $21, 300.

Thus, E [profit] = E [revenue] − E [cost] = $9, 300.

Problem 1: Matt Herron

Estimated demand	Probability
100	0.1
200	0.4
300	0.4
400	0.1

(1a) Calculate the expected profits for each of the following four cases: when Matt buys 100, 200, 300 or 400 overcoats. How many overcoats should Matt buy if he wishes to maximize expected profits over the coming Christmas season?

Q = 300: He orders 300, has a 0.10 probability of having an excess of 200 (when D = 100), has a 0.40 probability of having an excess of 100 (when D = 200), and sells all ordered units with probability 0.50 (when D = 300 or 400).

E [cost] = 300 · 60 = \$18, 000.

E [revenue] = 0.10 · (100 · 110 + 200 · 40) + 0.40 · (200 · 110 + 100 · 40) + 0.50 · (300 · 110) = \$28, 800.

Thus, E [profit] = E [revenue] − E [cost] = \$10, 800.

Problem 1: Matt Herron

Estimated demand	Probability
100	0.1
200	0.4
300	0.4
400	0.1

(1a) Calculate the expected profits for each of the following four cases: when Matt buys 100, 200, 300 or 400 overcoats. How many overcoats should Matt buy if he wishes to maximize expected profits over the coming Christmas season?

Q = 400: He orders 400, has a 0.10 probability of having an excess of 300 (when D = 100), has a 0.40 probability of having an excess of 200 (when D = 200), a 0.40 probability of having an excess of 100 (when D = 300), and sells all orders with probability 0.10 (when D = 400).

E [cost] = 400 · 60 = $24, 000.

E [revenue] = 0.10 · (100 · 110 + 300 · 40) + 0.40 · (200 · 110 + 200 · 40) + 0.40 · (300 · 110 + 100 · 40) + 0.10 · (400 · 110) = $33, 500.

Thus, E [profit] = E [revenue] − E [cost] = $9, 500.

Thus, we find that the optimal (profit-maximizing) order quantity is $Q^* = 300$, with an expected profit equal to $10, 800.

Problem 1: Matt Herron

Estimated demand	Probability
100	0.1
200	0.4
300	0.4
400	0.1

(1b) Now solve (1a) again using marginal analysis.

We follow the steps in Recipe #3. The critical ratio is

$$\frac{C_u}{C_u + C_o} = \frac{r - c}{r - s} = \frac{110 - 60}{110 - 40} = \frac{50}{70} = 0.714$$

We are in the Case 2 of the Recipe: we have a discrete distribution of demand. But we don't know it; we are given the density function! Summing up step-by-step the probabilities we get the discrete distribution of demand:

Quantity (Q)	$F(Q) = P[D \leq Q]$
100	0.1
200	0.5
300	0.9
400	1

Now, we need to find a value for Q such that F (Q) = 0.714. This value is not included in the above table, so using the round-up from above rule of the Recipe, we find that the optimal order quantity is Q^* = 300, which corresponds to an expected profit of $10,800 (calculated above).

Problem 1: Matt Herron

(1c) The assumption of a discrete demand distribution is a bit artificial. Solve the problem using a normal distribution for demand (with the same mean and standard deviation as the discrete distribution of the original problem description). Does this change make much difference?

To use the Normal distribution, we first need to know its parameters, i.e. the mean (μ) and the variance σ^2. These would be identical to the mean and variance of the discrete distribution in (1a) and (1b).

Mean: $\mu = E[D] = 0.1 \cdot 100 + 0.40 \cdot 200 + 0.40 \cdot 300 + 0.10 \cdot 400 = 250$
Variance: $\sigma^2 = E\left[D^2\right] - (E[D])^2$. Therefore:

$$\begin{aligned}\sigma^2 &= E\left[D^2\right] - (E[D])^2 \\ &= \left(0.1 \cdot 100^2 + 0.40 \cdot 200^2 + 0.40 \cdot 300^2 + 0.10 \cdot 400^2\right) \\ &\quad - (0.1 \cdot 100 + 0.40 \cdot 200 + 0.40 \cdot 300 + 0.10 \cdot 400)^2 \\ &= 69,000 - 250^2 = 6,500\end{aligned}$$

and the standard deviation equals:

$$\sigma = \sqrt{\sigma^2} = \sqrt{6,500} = 80.6$$

Problem 1: Matt Herron

(1c) The assumption of a discrete demand distribution is a bit artificial. Solve the problem using a normal distribution for demand (with the same mean and standard deviation as the discrete distribution of the original problem description). Does this change make much difference?

We therefore assume a normal (continuous) demand model, with the same mean and the same standard distribution as the above discrete distribution.

The underage and overage costs have not changed and therefore, we need to find z^* corresponding to the critical ratio 0.714. From Table 1 in your recipe #3, this is $z^* = 0.57$.

Hence, in the normal distribution case, we have that

$Q^* = \mu + z^* \cdot \sigma = 250 + 0.57 \cdot 80.6 = 296$ overcoats

The continuous and the discrete models have the same mean and variance, so it is not surprising that the results for the optimal order quantities are close too.

Problem 1: Matt Herron

(1c) The assumption of a discrete demand distribution is a bit artificial. Solve the problem using a normal distribution for demand (with the same mean and standard deviation as the discrete distribution of the original problem description). Does this change make much difference?

Next, we calculate the expected profit.

Use the z^*-value you found before to look up in the Standard Normal Loss Function Table 2 from recipe #3 the corresponding L(z)-value.

Expected lost sales = σ · L (0.57) = 80.6 · 0.17705 = 14.27 overcoats

Expected sales = μ − Expected lost sales = 250 − 14.27 = 235.73 overcoats

Expected leftover inventory = Q^* − Expected sales = 296 − 235.73 = 60.27 overcoats

Expected profit = [(r − c) * Expected sales] − [(c − s) * Expected leftover inventory] = [($50) * 235.73] − [($20) * 60.27] = $10,581

The result for the continuous model ($10, 581) is indeed close to that obtained for the discrete model (in part 1a): $10, 800.

Problem 2: Eloise's Pastry

Eloise is baking fresh pastry at several shops throughout the Leipzig area. Each pastry sells for 4€ and costs 1.80€ to make. Each one not sold at the end of the day is sold the next morning for 1€ as "yesterday's bargain pastry".

The daily demand is distributed as follows:

Demand	100	150	200	250	300	350	400
Probability	0.05	0.10	0.20	0.25	0.25	0.10	0.05

Problem 2: Eloise's Pastry

a. How many pastries should Eloise bake each day? How many of those will she expect to sell the same day, how many the next day, and what is her expected daily profit?

Follow the steps in the recipe sheet. Demand has a discrete distribution. We calculate the mean and variance as follows:

$$
\begin{aligned}
E(\text{Demand}) &= \mu = 0.05 * 100 + 0.10 * 150 + 0.20 * 200 + 0.25 * 250 + 0.25 * 300 + 0.10 * 350 + 0.05 * 400 \\
&= 252.5 \\
\sigma^2 &= 0.05(100 - 252.2)^2 + 0.10(150 - 252.5)^2 + 0.20(200 - 252.5)^2 + 0.25(250 - 252.5)^2 + \\
&+ 0.25(300 - 252.5)^2 + 0.10(350 - 252.5)^2 + 0.05(400 - 252.5)^2 \\
&= 5368.75 \\
\sigma &= \sqrt{\sigma^2} = 73.3
\end{aligned}
$$

c = €1.80, r = €4.00, s = €1.00.

Hence we get: $C_u = r - c = 4 - 1.80 = 2.20$

$C_o = c - s = 1.80 - 1 = 0.80$

Problem 2: Eloise's Pastry

a. How many pastries should Eloise bake each day? How many of those will she expect to sell the same day, how many the next day, and what is her expected daily profit?

Critical ratio:

$$\frac{C_u}{C_u + C_o} = \frac{2.2}{2.2 + 0.8} = 0.733$$

We want to find the order quantity that gives F(Q^*) = P(Demand ≤ Q^*) = 0.733:

Demand	Probability	$F(Q^*)$
100	0.05	0.05
150	0.10	$0.10 + 0.05 = 0.15$
200	0.20	$0.20 + 0.15 = 0.35$
250	0.25	$0.25 + 0.35 = 0.60$
300	0.25	$0.25 + 0.60 = 0.85$
350	0.10	$0.85 + 0.10 = 0.95$
400	0.05	$0.05 + 0.95 = 1$

We have F(250) = 0.60 and F(300) = 0.85. Our critical ratio lies between these values. We choose the one closest from above, therefore: Q^* = 300.

Problem 2: Eloise's Pastry

a. How many pastries should Eloise bake each day? How many of those will she expect to sell the same day, how many the next day, and what is her expected daily profit?

Expected same-day and next-day sales:

E(Sales) can be calculated directly, considering each demand value and probability:

$E(\text{Sales}) = 0.05 * 100 + 0.1 * 150 + 0.2 * 200 + 0.25 * 250 + (0.25 + 0.10 + 0.05) * 300 = 242.5$

Note that if the demand is greater than or equal to 300, you sell 300 pastries.

The expected next-day sales is given by the remaining items in your inventory if you have demand less than 300 items (assume there's always enough demand for the "yesterday's bargain pastry"):

$E(\text{Leftover Inventory}) = 0.05 * (300 - 100) + 0.1 * (300 - 150) + 0.2 * (300 - 200) + 0.25 * (300 - 250) = 57.5$

Problem 2: Eloise's Pastry

a. How many pastries should Eloise bake each day? How many of those will she expect to sell the same day, how many the next day, and what is her expected daily profit?

Expected Profit:

Use the formula:

$E(\text{Profit}) = C_u * E(\text{Sales}) - C_o * E(\text{Leftover Inventory}) = 2.20 * 242.5 - 0.80 * 57.5 = 487.5$

Problem 2: Eloise's Pastry

b. Resolve part a, this time using a normal distribution for daily demand, with the same mean and variance as in part a.

Now the demand is normally distributed with $\mu = 252.5$ and $\sigma = 73.3$.

The critical ratio is 0.733. In the table for standard normal distribution we have 0.73237 for z = 0.62 and 0.73565 for $z^* = 0.63$. We choose $z^* = 0.63$.

The optimal order quantity is: $Q^* = \mu + \sigma * z^* = 252.5 + 0.63 * 73.3 = 299$ pastries (approx.). The answer is not that different from the discrete case.

To calculate the sales, leftover inventory and profits, we follow the steps:

E(Lost Sales) = $\sigma * L(z^*) = 73.33 * L(0.63) = 73.33 * 0.16059 = 11.77$ (Table 2)

E(Sales) = μ − E(Lost Sales) = 252.5 − 11.77 = 240.73

E(Leftover Inventory) = Q^* − E(Sales) = 299 − 240.73 = 58.27

E(Profits) = C_u ∗ E(Sales) − C_o ∗ E(Leftover Inventory) = 2.2 ∗ 240.73 − 0.8 ∗ 58.27 = 483

The results we get for the discrete and continuous cases are similar, so the continuous case is a good approximation for the discrete distribution.

Problem 3: Selection of electricity generating capacity

You are considering the decision on the quantity and type of electricity generating capacity to install for meeting the uncertain demand during the next quarter. Once chosen, this capacity remains fixed for the future.

Demand over this period is uncertain with a mean weekly demand rate of 1,000,000 units (of electricity) per week and a standard deviation of weekly demand of 200,000 units. To play it safe, you have decided to install sufficient capacity to be able to supply 1,600,000 units of electricity per week. This amount would indeed cover most reasonable demand scenarios (up to 3 standard deviations above the mean.)

The question you are facing concerns the mix of plants that will make up your capacity. You have two types of plants: base load and peak load plants. So the sum of the capacities of your base load and peak load plants should be 1,600,000 units of electricity per week. Base plants are more expensive to install, but cheaper to operate. Peak load plants have the opposite characteristic: they are more expensive to operate, but require a smaller capital outlay to bring online.

Problem 3: Selection of electricity generating capacity

The weekly capital cost for a unit of base capacity is 3.3€, that of a unit peak load capacity is only 2.5€. Capital costs are to be paid regardless of usage. Operating costs are to be paid only as a function of usage. The estimated operating cost for delivering a unit of electricity in a given week with base load is equal to 1.7€, whereas the estimated cost of using peak load is equal to 4€ per unit delivered.

a. What is, given the above parameters, the optimal mix of peak and base load capacity?

The main point is to understand the concept of overage and underage costs.

On the one hand, it is cheaper to supply one unit of electricity using the base load plant: you pay 3.3 + 1.7 = 5€ for a unit of base load capacity, and 2.5 + 4.0 = 6.5€ for a unit of peak load capacity. Therefore, you lose money if you haven't installed a unit of base load capacity and you have demand for an extra unit of electricity. The underage cost is the profit margin that the firm loses by not having a unit of base load capacity, given that there is demand for an extra unit. This means that we satisfy that unit of demand from the peak load capacity instead. Hence the underage cost: $C_u = 6.5 - 5 = 1.5$€

Problem 3: Selection of electricity generating capacity

The weekly capital cost for a unit of base capacity is 3.3€, that of a unit peak load capacity is only 2.5€. Capital costs are to be paid regardless of usage. Operating costs are to be paid only as a function of usage. The estimated operating cost for delivering a unit of electricity in a given week with base load is equal to 1.7€, whereas the estimated cost of using peak load is equal to 4€ per unit delivered.

a. What is, given the above parameters, the optimal mix of peak and base load capacity?

On the other hand, the base plant costs you too much to install. Therefore, you'll lose money if you build too much of the base plant and your demand turns out to be low. The overage cost is the cost that the firm incurs when there is no demand for a unit capacity. For a unit of base load capacity, the firm has to pay the capital cost 3.3€, whereas if the firm had a unit of peak load capacity instead, the capital cost would have been 2.5€. It follows that: $C_o = 3.3 - 2.5 = 0.8$€.

Problem 3: Selection of electricity generating capacity

The weekly capital cost for a unit of base capacity is 3.3€, that of a unit peak load capacity is only 2.5€. Capital costs are to be paid regardless of usage. Operating costs are to be paid only as a function of usage. The estimated operating cost for delivering a unit of electricity in a given week with base load is equal to 1.7€, whereas the estimated cost of using peak load is equal to 4€ per unit delivered.

a. What is, given the above parameters, the optimal mix of peak and base load capacity?

Then we have the critical ratio: $C_u/(C_u + C_o) = 1.5/(1.5 + 0.8) = 0.652$.

Using Table 1, $z^* = 0.40$. The optimal amount of base load capacity is: $Q^* = \mu + \sigma * z^* = 1,000,000 + 0,40 * (200,000) = 1,080,000$ units/week.

Finally, the optimal peak load capacity is: 1,600,000 - 1,080,000 = 520,000 units/week.

Problem 3: Selection of electricity generating capacity

b. What is the expected weekly cost of electricity supply given this mix?

Once you have 1,080,000 units of base load and 520,000 units of peak load installed, you would first operate the base load plants before the peak load plants (since it's cheaper to operate). For instance, if your demand turns out to be 800,000 you would satisfy it only using the base load capacity. The demand not satisfied using base load capacity will be satisfied by peak load capacity. Therefore, "expected lost sales" is actually the amount of use of peak load capacity. We then have (using Table 2):

$E(\text{Peak Load Sales}) = E(\text{Lost Sales}) = \sigma * L(z^*) = \sigma * L(0.40) = 200,000 * 0.23044 = 46,088$ units/week

$E(\text{Base Load Sales}) = E(\text{Demand}) - E(\text{Peak Load Sales}) = 1,000,000 - 46,088 = 953,912$ units/week

$E(\text{Operating Costs}) = 4.0 * 46,088 + 1.7 * 953,912 = 1,806,002$ €/week

$E(\text{Capital Costs}) = 3.3 * 1,080,000 + 2.5 * 520,000 = 4,864,000$ €/week

Total costs (capital + operating) = 6,670,002 €/week.

Problem 4: Storewars

Consider the situation by a manufacturer of playstations, SUNNY, who is facing Christmas demand. SUNNY sells the playstations through a large retail chain, TOYS-are-MINE, at a manufacturer (wholesale) price of $500 per playstation. SUNNY's unit manufacturing cost is $ 350 per playstation. TOYS-are-MINE offers the playstations for sale at its retail stores at the retail price of $800 per station. Its demand forecast for the upcoming Christmas season is for 200,000 playstations, with a standard deviation of 50,000. The retail chain has to order ahead of the season. Any playstations ordered will be delivered in time for the Christmas sales season, but no re-supply is possible during sales season.

Any playstation that will not be sold during the upcoming Christmas season will be sold after the season at a discount, for a price of $300 per station.

Problem 4: Storewars

(2a.i) What is the optimal order quantity for TOYS-are-MINE, and what are its expected profits on this item?

We follow the steps in Recipe #3. The critical ratio is

$$\frac{C_u}{C_u + C_o} = \frac{r - c}{r - s} = \frac{800 - 500}{800 - 300} = \frac{300}{500} = 0.60$$

We search for a z-value such that Φ(z)= 0.60.

Looking in Table 1 we find that the critical ratio value falls between two entries on the table z_1 = 0.25 and z_2 = 0.26.

Applying the round-up rule (closest from above), we choose the entry with the larger quantity, i.e. z^* = 0.26.

Therefore, Q^* = μ + z^* · σ = 200, 000 + 0.26 · 50, 000 = 213, 000 PlayStations.

Problem 4: Storewars

(2a.i) What is the optimal order quantity for TOYS-are-MINE, and what are its expected profits on this item?

Next, we calculate the expected profit of the retailer TOYS-are-MINE (TaM).

Use the z-value you found before to look up in the Standard Normal Loss Function Table 2 the corresponding L(z)-value.

TaM's Expected lost sales = $\sigma \cdot L(z^*) = 50,000 \cdot 0.28235 = 14,117.5$ PlayStations

TaM's Expected sales = μ − Expected lost sales = 200,000 − 14,117.5 = 185,882.5 PlayStations

TaM's Expected leftover inventory = Q^* − Expected sales = 213, 000 − 185,882.5 = 27,117.5 PlayStations

TaM's Expected profit = [(r − c) * Exp. sales] − [(c − s) * Exp. leftover inv.] = [($300) * 185,882.5] − [($200) * 27,117.5] = $50,341,250

Problem 4: Storewars

(2a.ii) What is the expected profit of the manufacturer, SUNNY?

SUNNY's profit from selling the PlayStation at TOYS-are-MINE store is 213, 000 * $150 = $31, 950, 000, where 213, 000 is the number of PlayStations that TOYS-are-MINE purchases and $150 is SUNNY's gross margin $500 − $350 = $150.

While TOYS-are-MINE might be quite pleased with this situation (it does earn $50m relative to SUNNY's $32m approximately), it should not be.

The total supply chain's profit is $31, 950, 000 + $50, 341, 250 = $82, 291, 250, but it could be higher as we explain next.

Problem 4: Storewars

(2b) What is the optimal order quantity on this item, if the retailer and the manufacturer were integrated into a single company?

Here we need to find the supply chain's optimal quantity, since this is the quantity that maximizes the integrated supply chain.

We can still use the newsvendor model to evaluate the supply chain's order quantity decision and performance measures.

Each lost sale costs the supply chain (SC) the difference between the retail price and the manufacturing cost, \$800 − \$350 = \$450; that is the SC's underage cost is C_u = \$450.

Each leftover PlayStation costs the SC the difference between the manufacturing cost and the salvage value, \$350 − \$300 = \$50; that is, the SC's overage cost is C_o = \$50.

The SC's critical ratio is

$$\frac{C_u}{C_u + C_o} = \frac{450}{450 + 50} = 0.9$$

Problem 4: Storewars

(2b) What is the optimal order quantity on this item, if the retailer and the manufacturer were integrated into a single company?

The appropriate z-statistic for that critical ratio is $z^* = 1.29$, because $\Phi(1.28) = 0.89973$ and $\Phi(1.29) = 0.90147$.

The SC's expected profit-maximizing order quantity is then

$Q^* = \mu + z^* \cdot \sigma = 200,000 + 1.29 \cdot 50,000 = 264,500$

which is higher than TOYS-are-MINE's order of 213, 000 PlayStations.

Problem 4: Storewars

(2c) Compare the expected profits generated by the integrated supply chain (which you computed in your answer to (2b)), with that of the chain formed by the two independent firms, SUNNY and TOYS-are-MINE. Which arrangement generates the highest expected profits for the supply chain? Why? What other major differences do you observe between these two inter-organizational arrangements?

The SC's performance measures can then be evaluated assuming the SC optimal order quantity, 264, 500 units.

SC's Expected lost sales = $\sigma \cdot L(1.29) = 50,000 \cdot 0.0465 = 2,325$ PlayStations

SC's Expected sales = μ − Expected lost sales = 200, 000 − 2, 325 = 197, 675 PlayStations

SC's Expected leftover inventory = Q^* − Expected sales = 264, 500 − 197, 675 = 66, 825 PlayStations

SC's Expected profit = [(r − c) * Exp. sales] − [(c − s) * Exp. leftover inv.] = [($450) * 197, 675] − [($50) * 66, 825] = $85, 612, 500

Problem 4: Storewars

(2c) Compare the expected profits generated by the integrated supply chain (which you computed in your answer to (2b)), with that of the chain formed by the two independent firms, SUNNY and TOYS-are-MINE. Which arrangement generates the highest expected profits for the supply chain? Why? What other major differences do you observe between these two inter-organizational arrangements?

Thus, while SUNNY and TOYS-are-MINE currently earn an expected profit of $82,291,250, their SC could enjoy an expected profit that is about 4 percent higher, or $3,321,250.

But, why the current supply chain perform worse than it could? The obvious answer is that TOYS-are-MINE does not order enough PlayStations: TOYS-are-MINE orders 213, 000 of them, but the SC's optimal order quantity is 264, 500 units.

But why doesn't TOYS-are-MINE order enough? Because TOYS-are-MINE is acting in its own self interest to maximize its own profit.

Problem 4: Storewars

(2c) Compare the expected profits generated by the integrated supply chain (which you computed in your answer to (2b)), with that of the chain formed by the two independent firms, SUNNY and TOYS-are-MINE. Which arrangement generates the highest expected profits for the supply chain? Why? What other major differences do you observe between these two inter-organizational arrangements?

To explain further, TOYS-are-MINE must pay SUNNY $500 per PlayStation and so TOYS-are-MINE acts as if the cost to produce each PlayStation is $500, not the actual $350.

From TOYS-are-MINE's perspective, it does not matter if the actual manufacturing cost is $350, $400, or even $0; its "manufacturing cost" is $500.

This is related to an important phenomenon called double marginalization. In simple words, TOYS-are-MINE only earns a "piece of the pie" ($300) of the total "pie" benefit per sale ($450). Thus, it is logical that TOYS-are-MINE is not willing to purchase as much inventory as would be optimal for the supply chain.